

Cultural Homophily in Grand Slam Doubles

ATP Men's Grand Slam Doubles, 2018–2025 · 1,793 matches after dropping 47 retirements / walkovers · Regression sample: 3,432 team-obs (complete ranking, nationality/language for all 4 players; GS experience imputed to 0 for first-timers)

1. Data and Measures

Three culture measures capture teammate similarity. **Same nationality** and **same official language** are binary (0/1). **Language proximity** is a continuous 0–1 index of ethnolinguistic closeness (higher = more similar language family).

Regressions use a team-level panel with tournament-by-year fixed effects, round fixed effects, and standard errors clustered by match. Controls in all main specifications: team average doubles ranking, opponent average ranking, top-100 singles indicator (either player), team average GS doubles appearances (*exp_mean*), and its square. Culture measures enter one at a time. Rank gap between teammates was tested as an additional control and is insignificant throughout; it is excluded from all main specifications. Nationality and language data are fully resolved in the pipeline (birthplace fallback, surname lookup, Monaco language correction, and `manual_nationality.csv`); zero matches are dropped for nationality.

2. Observation Breakdown

Step	Description	Matches	Change	Team-obs
1	Raw scraped dataset (2018–2025, excl. 2020)	1,840	—	3,680
2	Drop retirements / walkovers (20 walkovers · 9 retired in set 1 · 18 retired in set 2+)	1,793	–47	3,586
3	Drop: doubles ranking incomplete for ≥1 player	1,779	–14	3,558
4	Drop: nationality / language missing for ≥1 player (pipeline fully resolves: birthplace, surname lookup, Monaco fix, <code>manual_nationality.csv</code>)	1,779	0	3,558
5	Exclude Olympic matches (Tokyo 2021 · Paris 2024)	1,716	–63	3,432
6	GS experience: 210 obs are first-time GS participants (no prior appearances); <i>exp_mean</i> imputed to 0 — no obs dropped (these teams have mean rank ≈193 vs ≈124 for the rest; dropping would create selection bias)	—	0	3,432
→ Grand Slams regression sample (Tables 3–4)		—		3,432

Table 4 (tiebreak win): 940 tiebreaks × 2 team-obs = 1,880 team-obs (747 unique matches).

3. Pressure Outcome Counts

Outcome	N	Share of matches
Set-1 tiebreak (7-pt)	445	24.8%
Set-2 tiebreak (7-pt)	470	26.2%
Any regular tiebreak (sets 1–2)	804	44.8%
Match tiebreak / super-tb (10-pt, set 3)	24	1.3%
Any tiebreak (all types)	896	50.0%
Match went to 3 sets	787	43.9%
Comeback wins (winner lost set 1)	351	19.6%

All matches (GS + Olympics, N=1,793). Match tiebreaks concentrated in Olympics (N=20) and Wimbledon post-2018 (N=4).

4. Olympic Cycle Descriptive Evidence

Table 2A — Tokyo 2021 cycle (Grand Slams only, Olympics excluded)

Period	GS window	N teams	Same Nat.	Diff Nat.	Same Lang.	Diff Lang.	Ling. Prox.	Diff Prox.
Pre-Tokyo	AO18 – USO19	1,006	40.1%	59.9%	55.8%	44.2%	56.7%	43.3%
Tokyo Prep	AO20 – Wim21 (ex Wim20)	364	42.0%	58.0%	57.0%	43.0%	57.0%	43.0%
Post-Tokyo	USO21 – USO22	604	39.2%	60.8%	55.4%	44.6%	56.3%	43.7%

N teams = GS matches × 2. Same/Diff = share of all teams (pooled winner + loser) on each cultural dimension.

Table 2B — Paris 2024 cycle (Grand Slams only, Olympics excluded)

Period	GS window	N teams	Same Nat.	Diff Nat.	Same Lang.	Diff Lang.	Ling. Prox.	Diff Prox.
Pre-Paris	AO22 – USO22	486	39.5%	60.5%	54.8%	45.2%	55.9%	44.1%
Paris Prep	AO23 – Wim24	860	42.6%	57.4%	58.5%	41.5%	58.9%	41.1%
Post-Paris	USO24 – USO25	626	43.9%	56.1%	58.9%	41.1%	60.4%	39.6%

N teams = GS matches × 2.

Cultural Homophily in Grand Slam Doubles - Tokyo 2021 Cycle
(Same vs. Diff, by Tournament-Year)

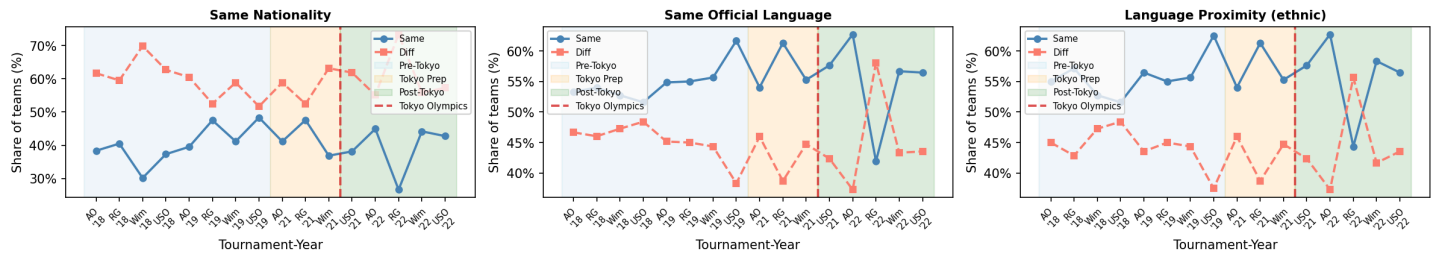


Figure 1. Share of same- vs. different-culture teams by tournament-year, Grand Slams only. Shaded: Pre-Tokyo (blue), Tokyo Prep (orange), Post-Tokyo (green). Red dashed = Tokyo Olympics 2021.

Cultural Homophily in Grand Slam Doubles - Paris 2024 Cycle
(Same vs. Diff, by Tournament-Year)

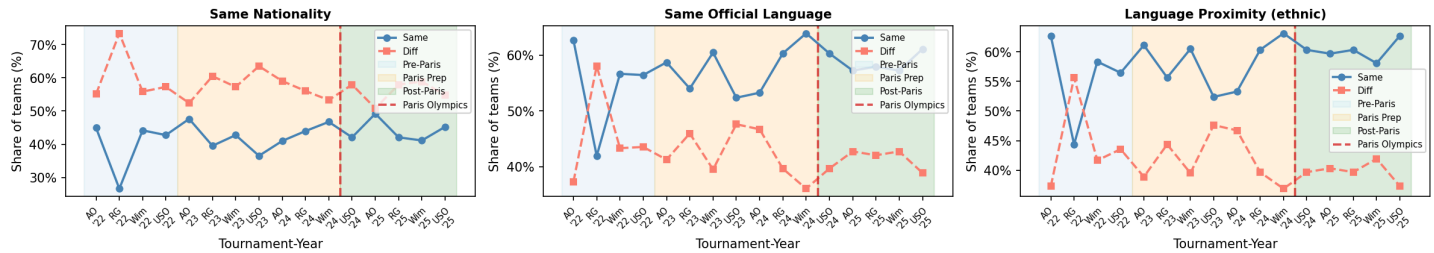


Figure 2. Paris 2024 cycle. Red dashed = Paris Olympics 2024.

Cultural Homophily in Grand Slam Doubles -- Full Timeline 2018-2025
(Tokyo & Paris Olympic Cycles Combined, Same vs. Diff)

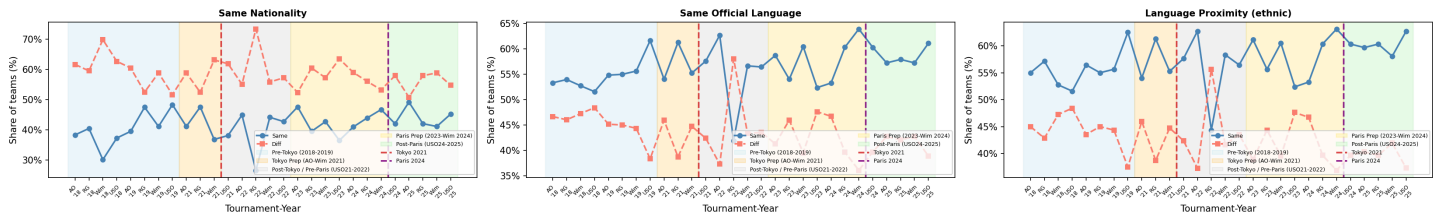


Figure 3. Both Olympic cycles on a single timeline (2018–2025). Vertical dashed lines mark Tokyo 2021 and Paris 2024.

5. Regression Results

Average marginal effects (AME, dP/dx) from logit models. Each column is a separate model. Standard errors in parentheses, clustered by match. FE: tournament $\times$ year, round. All specifications include GS experience controls (*exp_mean* and its square). Culture measures enter one at a time.

Table 3 — Outcome: Match Win | N = 3,432 team-obs | AME (dP/dx)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.0343 ^{**} (0.0170) p=0.044	+0.0413 ^{**} (0.0164) p=0.012	+0.0379 ^{**} (0.0164) p=0.021
Team avg. doubles rank	−0.0007 ^{***} (0.0001)	−0.0007 ^{***} (0.0001)	−0.0007 ^{***} (0.0001)
Opponent avg. doubles rank	+0.0007 ^{***} (0.0001)	+0.0007 ^{***} (0.0001)	+0.0007 ^{***} (0.0001)
Top-100 singles player	−0.0035 (0.0323) p=0.914	−0.0019 (0.0324) p=0.953	−0.0014 (0.0323) p=0.966
Team avg GS appearances	+0.0184 ^{***} (0.0052) p=0.000	+0.0182 ^{***} (0.0052) p=0.000	+0.0180 ^{***} (0.0052) p=0.001
GS experience squared	−0.0006 ^{***} (0.0002) p=0.009	−0.0006 ^{***} (0.0002) p=0.009	−0.0006 ^{**} (0.0002) p=0.010
Pseudo-R ²	0.098	0.098	0.098

*** p<0.01 ** p<0.05 * p<0.10. Clustered SE. FE: tournament×year, round. All three culture measures statistically significant at the 5% level. Experience shows a concave effect: positive linear (p<0.01), negative quadratic (p<0.01). Top-100 singles indicator is not significant: within-GS doubles, having a top singles player does not predict match win once ranking and experience are controlled.

Table 4 — Outcome: Tiebreak Win | Unit: team × tiebreak | N = 1,880 obs | AME (dP/dx)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.0176 (0.0258) p=0.494	+0.0049 (0.0248) p=0.842	+0.0002 (0.0247) p=0.993
Team avg. doubles rank	−0.0004 ^{***} (0.0001) p=0.000	−0.0003 ^{***} (0.0001) p=0.000	−0.0003 ^{***} (0.0001) p=0.000
Opponent avg. doubles rank	+0.0004 ^{***} (0.0001) p=0.000	+0.0004 ^{***} (0.0001) p=0.000	+0.0004 ^{***} (0.0001) p=0.000
Top-100 singles player	−0.0302 (0.0488) p=0.536	−0.0289 (0.0489) p=0.554	−0.0288 (0.0488) p=0.555
Team avg GS appearances	+0.0025 (0.0122) p=0.838	+0.0017 (0.0122) p=0.891	+0.0014 (0.0122) p=0.909
GS experience squared	+0.0000 (0.0004) p=0.989	+0.0000 (0.0004) p=0.948	+0.0000 (0.0004) p=0.932
Pseudo-R ²	0.018	0.018	0.018

Unit = one team per tiebreak (2 obs per tiebreak). Includes set-1 7-pt (376), set-2 7-pt (391), set-3 7-pt (170), and set-3 10-pt match TB (3). Controls: rank_mean, opp_rank_mean, single_top100, exp_mean, exp_mean_sq (same as Table 3). FE: tournament×year + round. SE clustered by match. Culture AMEs are small and insignificant across all three measures — homophily advantage does not extend to within-match tiebreak outcomes. Ranking controls remain strongly significant ($p < 0.001$), confirming that the model captures real quality differences; experience controls are not significant in the tiebreak sample.

6. Heterogeneity Analysis

Three sets of interactions are estimated on the main match-win sample ($N = 3,432$ team-obs): (a) culture × GS experience, testing whether experienced teams benefit more from cultural alignment; (b) culture × surface, testing surface-specific coordination demands; and (c) culture × Hofstede individualism score (team-level, demeaned), testing whether collectivist teams gain more from cultural homophily.

Hofstede individualism scores (IDV) are taken from Geert Hofstede's official 6-dimension dataset (2015 release, geerthofstede.com). Countries not covered by Hofstede are assigned proxy scores following two tiers: *Tier 1* (direct cultural/linguistic kin, high confidence): BIH→SRB, CYP→GRC, MCO→FRA, MDA→ROU; *Tier 2* (same-region / closest available, moderate confidence): BLR→RUS, BOL→PER, DOM→COL, LBN→Arab composite (IDV=38), TUN→MAR, GEO→TUR, KAZ→RUS, UKR→RUS, UZB→RUS. The team-level score is $IC_team = (IDV_p1 + IDV_p2) / 2$, demeaned across the GS regression sample (mean = 62.95, SD = 20.94).

Table 6a — Heterogeneity: Culture × GS Experience | N = 3,432

Main effect of culture evaluated at zero GS experience; interaction captures how the marginal effect changes per additional Grand Slam appearance.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C)	+0.0269	−0.0391	−0.0416
	(0.0420)	(0.0427)	(0.0426)
	p=0.522	p=0.359	p=0.329
C × exp_mean	+0.0006	+0.0066**	+0.0065**
	(0.0033)	(0.0032)	(0.0032)
	p=0.846	p=0.040	p=0.043
Pseudo-R ²	0.098	0.099	0.099
N	3,432	3,432	3,432

Controls: rank_mean, opp_rank_mean, single_top100, exp_mean, exp_mean_sq. FE: tournament×year + round. SE clustered by match. The negative main effect of culture in cols (2)–(3) is evaluated at zero GS experience (teams with no prior GS appearances); the positive and significant interaction implies the language/proximity advantage grows with experience: at the sample mean of exp_mean ≈ 12 appearances, the implied total marginal effect is approximately +0.039–0.040, consistent with Table 3. Nationality homophily shows no significant experience gradient (p=0.846).

Table 6b — Heterogeneity: Culture × Surface | N = 3,432

Hardcourt (Australian Open + US Open) is the omitted baseline. Clay = Roland Garros; Grass = Wimbledon. Note: since surface is perfectly determined by tournament (each GS is on one surface), the surface main effects may be absorbed by the tournament×year FE; the interaction terms capture whether the culture coefficient differs across surfaces.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C) — hardcourt baseline	+0.0396	+0.0402*	+0.0337
	(0.0244)	(0.0237)	(0.0238)
	p=0.102	p=0.087	p=0.148
C × clay	−0.0098	−0.0080	+0.0062
	(0.0416)	(0.0408)	(0.0409)
	p=0.806	p=0.839	p=0.876
C × grass	−0.0112	+0.0125	+0.0102
	(0.0421)	(0.0409)	(0.0409)
	p=0.784	p=0.752	p=0.796
Pseudo-R ²	0.098	0.098	0.098
N	3,432	3,432	3,432

All surface interactions insignificant (p > 0.75 throughout). The culture advantage documented in Table 3 does not vary significantly across playing surfaces.

Table 6c — Heterogeneity: Culture × Hofstede Individualism (demeaned) | N = 3,432

IC_team_dm = team-average Hofstede IDV score minus the sample mean (62.95). Positive values = more individualist than average. Hypothesis: more individualist teams rely less on in-group bonding → negative θ . Two specifications per culture measure are shown: Spec 1 excludes the main IC effect; Spec 2 includes it.

Variable	Same Nationality		Same Language		Lang. Proximity	
	Spec 1	Spec 2	Spec 1	Spec 2	Spec 1	Spec 2
Culture (C)	+0.0320 [*]	+0.0275	+0.0391 ^{**}	+0.0332 [*]	+0.0356 ^{**}	+0.0296 [*]
	(0.0177)	(0.0178)	(0.0169)	(0.0174)	(0.0169)	(0.0174)
	p=0.063	p=0.112	p=0.017	p=0.050	p=0.030	p=0.080
IC_team_dm	—	+0.0012	—	+0.0011	—	+0.0011
	—	(0.0006)	—	(0.0008)	—	(0.0008)
		p=0.054		p=0.164		p=0.166
C × IC_team_dm (θ)	+0.00053	−0.00061	+0.00066	−0.00042	+0.00069	−0.00039
	(0.00056)	(0.00084)	(0.00048)	(0.00093)	(0.00048)	(0.00093)
	p=0.340	p=0.456	p=0.167	p=0.645	p=0.149	p=0.671
Pseudo-R ²	0.098	0.099	0.098	0.099	0.098	0.099
N		3,432		3,432		3,432

Controls: rank_mean, opp_rank_mean, single_top100, exp_mean, exp_mean_sq. FE: tournament×year + round. SE clustered by match. The culture main effects in Spec 1 are consistent with Table 3. The interaction θ is small and insignificant across all specifications ($p > 0.14$), providing no evidence that individualism moderates the cultural homophily advantage. In Spec 2 the main IC_team_dm coefficient is positive for same nationality ($p=0.054$, marginal) — teams with higher individualism scores may perform marginally better — but this is insignificant for the language measures ($p>0.16$).